

Emerging patterns of somatic mutations in cancer

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Abstract | Recent advances in technological tools for massively parallel, high-throughput sequencing of DNA have enabled the comprehensive characterization of somatic mutations in a large number of tumour samples. In this Review, we describe recent cancer genomic studies that have assembled emerging views of the landscapes of somatic mutations through deep-sequencing analyses of the coding exomes and whole genomes in various cancer types. We discuss the comparative genomics of different cancers, including mutation rates and spectra, as well as the roles of environmental insults that influence these processes. We highlight the developing statistical approaches that are used to identify significantly mutated genes, and discuss the emerging biological and clinical insights from such analyses, as well as the future challenges of translating these genomic data into clinical impacts.

Next-generation sequencing (NGS). All post-Sanger DNA sequencing methods, most commonly referring to massively parallel sequencing technology.

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Although it is recognized that cancer is a collection of complex pathological entities with diverse biological capabilities¹, much of our current understanding of cancer genetics is grounded on the principle that cancer arises from a clone that has accumulated the requisite somatically acquired genetic aberrations, leading to malignant transformation². The progression of such a transformed clone to disseminated disease was previously thought to be a linear process driven by serially acquired new mutations, including base pair substitutions, small insertions and deletions (indels) of bases, chromosomal rearrangements, and gains and losses in gene copy number³. However, recent insights have emerged from comprehensive genomic characterization using next-generation sequencing (NGS) technologies, which have allowed the sequencing of the coding portion of the genome through hybrid-capture, whole-exome sequencing (WES) or of nearly all base pairs in a tumour–normal pair by whole-genome sequencing (WGS) (BOX 1) (reviewed in REFS 4,5), revealing previously unanticipated complexities in the patterns of somatic alterations in cancers.

Before the widespread use of NGS technologies, studies using PCR amplification and Sanger-based capillary sequencing methodology were limited by cost and throughput, dictating either one of two study designs: either a limited number of genes sequenced in a large cohort or all coding genes sequenced in a small number of samples. Mutations in many of the known cancer genes, so-called ‘mountains’ (genes that are altered in

a large proportion of tumours), were discovered by focused Sanger-based sequencing or by cytogenetic analyses; some of these genes express mutated proteins that have become successful drug targets^{2,4}. The first efforts to interrogate the complete protein-coding sequence of the genomes of colon cancer, breast cancer and glioblastoma necessitated the generation of up to 208,311 primer pairs^{6,7} for Sanger-based sequencing. These pioneering studies led to the identification of a highly recurrent mutation in a novel oncogene, isocitrate dehydrogenase 1 (*IDH1*)⁸, which is involved in both cell metabolism^{8,9} and DNA methylation¹⁰, reinforcing the promise of unbiased genomic sequencing in the identification of novel genetic driving events in human cancers.

The recent advances in NGS technology have allowed the cancer research community to use systematic sequencing to identify additional mountains, which include frequent mutations in epigenetic regulators and pre-mRNA splicing machinery in many cancers (FIG. 1) (described in detail below), and so-called ‘hills’ (genes that are altered less frequently in cancer). Furthermore, these efforts have uncovered various mutational mechanisms underlying tumorigenesis and progression, such as chromothripsis^{11,12}, chromoplexy^{13,14} and kataegis^{15–17}. Following the launch of the [Cancer Genome Project](#) (CGP) in the United Kingdom in 2000 and [The Cancer Genome Atlas](#) (TCGA) in the United States in 2006, the [International Cancer Genome Consortium](#) (ICGC) was created in 2007 to coordinate the generation

Hybrid-capture

A target enrichment approach in which custom oligonucleotides (the bait set) are designed and optimized to hybridize to specific regions of the genome so that specific fragments of DNA can be enriched by hybridization for next-generation sequencing.

Whole-exome sequencing

(WES). Next-generation sequencing of all protein-coding exomes after capture, using hybridization to a whole-exome bait set designed to enrich DNA in all protein-coding portions of the genome. The most common implementation also targets microRNA genes. The size of the captured DNA is approximately 40 Mb.

Whole-genome sequencing

(WGS). Sequencing of the entire genome, usually by random fragmentation (shotgun) and to sufficient coverage so as to ensure adequate representation of all alleles. A variation specifically using low-coverage WGS is sometimes used to assess rearrangements in the genome.

Cancer genes

Genes that harbour cancer-driving genetic aberrations (however, cancer genes may possess both driver and passenger somatic alterations) as defined by criteria including statistical evidence of selection and recurrence pattern, or by functional activity.

Chromothripsis

Greek for chromosome 'shattering', in which up to hundreds of genomic rearrangements take place in a single cellular crisis event that develops from errors in mitosis that occur in ~2–3% of cancers.

Chromoplexy

Greek for chromosome 'weave' or 'braid'. A process whereby a closed chain of chromosomes is formed by copy-neutral rearrangements that consist of 4–12 distinct breakpoint junctions, which tend to occur at transcriptionally active portions of chromatin.

of comprehensive catalogues of genome alterations from 52 different cancer types¹⁸. Currently, disease working groups from the TCGA and the ICGC have released comprehensive genomic analyses for various cancer types^{19–33}. In addition, a major ongoing effort is to molecularly characterize rare malignancies and cancers that are common in diverse geographical regions in different populations, such as gastric cancer^{34,35}, viral hepatitis (B or C)-associated hepatocellular carcinoma (HCC)^{27,36–38} and parasite-induced cholangiocarcinoma (CCA)³⁹.

The general approach to sequencing and bioinformatics analysis, as well as the potential of carrying out large-scale cancer genomics studies using NGS, has been discussed in depth^{2,4,5,40,41}. In this Review, we attempt to keep pace with the explosion in the number of NGS studies by summarizing a selection of findings from recently published genomic studies (from 2008 to July 2013), with a focus on the analysis of genes that are targeted by point mutations (base pair substitutions) and indels. We review WES and WGS studies that have provided the clearest picture of the landscape of somatic mutations in major adult tumour types (most of these studies involve deep-sequencing analyses of the coding exomes and the whole genomes of 20 or more samples per study), publications by the TCGA and the ICGC, as well as a few examples of rare malignancies and cancers that are more prevalent in diverse geographical regions (TABLE 1). We discuss our current understanding of the mutational landscapes of these different tumour types and underscore the biological insights into the aetiology of cancer gained through integrative and pathway analyses using various genomic platforms, while highlighting the challenges of identifying driver mutations and functionally validating significantly mutated genes (SMGs). Although structural rearrangement can also be identified through NGS, the analytical tools for defining such structural alterations are much less mature, thus we do not cover this important aspect of genomic alterations in detail.

Features of cancer genome studies

Mutation rates and spectra across cancer types. A meta-analysis of 2,957 whole exomes and 126 whole

genomes from 27 cancer types, which was carried out at the Broad Institute, USA, recently illustrated the mutational heterogeneity in diverse cancer types⁴². First, the variation in mutation frequency can be partly explained by cancer type. For example, paediatric and haematological cancers have the lowest mutation rates (approximately one mutation per megabase for chronic lymphocytic leukaemia (CLL))³⁰, compared with cancers in which environmental mutagens are known to increase the mutation burden, such as melanoma and lung cancer (~15 mutations per megabase for melanoma)⁴³. In addition, mutation rates can vary tremendously within a cancer type, often owing to the degree of exposure to an environmental mutagen, or dependent on the particular genes that are mutated (for example, tumours with mutations in mismatch repair genes will have higher mutation rates). Second, the mutation spectra also vary across cancer types. For example, clustering analysis on all possible mutations (considering the context of flanking residues) demonstrated natural groupings of mutation spectra and cancer types that are consistent with known signatures of carcinogenesis mechanisms: lung tumours have a high proportion of G→T transversions, which is attributable to exposure to polycyclic aromatic hydrocarbons from tobacco smoke; melanomas have a high proportion of C→T transitions in dipyrimidines caused by ultraviolet (UV) radiation-induced DNA damage and misrepair; gastrointestinal (oesophageal, colorectal and gastric) tumours have a high frequency of transition mutations at CpG dinucleotides that may be a reflection of increased methylation levels in these tumours²²; cervical, bladder, some head and neck, and breast cancers frequently have mutations at cytosines in the context of TpC dinucleotides, which are characteristic mutations caused by the APOBEC family of cytidine deaminases^{15–17}; and leukaemias (acute myeloid leukaemia (AML) and CLL) have A→T mutations in the context of TpA dinucleotides⁴². Finally, WGS analysis confirmed that mutational heterogeneity across the genome is heavily influenced by two factors: gene expression level and DNA replication timing. In light of this known mutational heterogeneity, correcting for such factors has been shown to be important for determining SMGs^{42,43}.

Box 1 | Whole-exome and whole-genome sequencing: complementarity and pitfalls

To identify mutated genes that are relevant to cancer, the first step is to generate high-quality next-generation sequencing data covering the genomic regions of interest (for example, whole-genome sequencing (WGS) or whole-exome sequencing (WES), or a targeted panel of selected genes) from both the germline DNA and tumour DNA. Next, the 'raw' reads from the sequencers need to be mapped back to the reference human genome so that they can be properly 'stitched' together to reconstruct the linear DNA sequence that corresponds to each chromosome. Once aligned, sophisticated analytical algorithms are used to call sequence variants in the tumour DNA compared with the human reference genome and to determine whether a variant is 'somatic' by comparison with the germline DNA. Deep coverage (most WES and WGS studies aim for an average of 100- to 150-fold and 30- to 60-fold coverage, respectively) is necessary owing to the heterogeneous nature of most tumours^{4,5,40,41}. Tumours can consist of many subclones¹²⁹, can have gross numerical and structural chromosome abnormalities resulting in large variation in actual DNA content (ploidy)¹³⁰, and often reside in a community of 'genomically normal' non-cancerous cells — all of which reduce the sensitivity to detect mutations. This can be further exacerbated in genomic regions of high GC content, which typically results in low to absent coverage. Therefore, it is important that the analytical tools that are used to call a somatic mutation consider whether the genomic locus is adequately covered.

Approaches to identify cancer-associated genes. Identifying which mutations are likely to be drivers in pathogenesis and elucidating how the mutated genes affect the biology of a given tumour are fundamental challenges in cancer genomics. Statistical tools have been developed by various groups to identify SMGs with a higher mutation rate than the expected (calculated or

estimated) background mutation rate (BMR), which indicates positive selection during tumorigenesis (BOX 2). Following the recent surge of WES and WGS studies, a more comprehensive ‘census’ of human cancer genes has emerged. However, this is only the first step in the translation of these findings into clinical benefits for patients. Additionally, many published studies have collected data on other genomic dimensions (for example, DNA copy number, DNA methylation, and mRNA and microRNA (miRNA) expression profiles) that can be used for integrative analysis (BOX 3) and for elucidating the mechanisms of disease pathogenesis; we highlight a few examples of such integration in the sections below on specific cancers.

Landscape of somatic mutations in solid tumours

Glioblastoma. The Sanger-based sequencing of 20,661 protein-coding genes from 22 glioblastoma samples mentioned above revealed a recurrent heterozygous *IDH1* mutation targeting arginine 132 in 12% of samples, which correlated with improved patient survival⁶. In parallel, glioblastoma was the first cancer type to undergo comprehensive genomic characterization by the TCGA Research Network. A targeted approach consisting of Sanger-based capillary sequencing of 601 selected genes in 91 glioblastoma–normal pairs was used to identify somatic mutations, which revealed frequent mutations in the phosphatidylinositol 3-kinase (PI3K) regulatory subunit *PIK3R1* (REF. 24). The targeted sequencing was complemented by analyses of DNA copy number, mRNA expression and DNA methylation from 206 glioblastoma samples. Integrative analysis demonstrated significant deregulation of the receptor tyrosine kinase (RTK)–RAS–PI3K, p53 and retinoblastoma (RB) signalling pathways in glioblastoma²⁴. This study set the framework for comprehensive multi-dimensional genomic characterization in large-scale studies (BOX 3). Investigation of RNA sequencing (RNA-seq) data from glioblastomas also revealed translocations involving fibroblast growth factor receptor (*FGFR*) genes becoming fused to the transforming acidic coiled-coil (TACC) coding domains of *TACC1* or *TACC3* genes in approximately 3% of cases (three of 97)⁴⁴. Glioblastoma cells expressing the resulting fusions demonstrated sensitivity to FGFR kinase inhibition in preclinical assays⁴⁴. The examples mentioned above demonstrate the power of unbiased genomic sequencing in the identification of clinically relevant genetic changes.

Clear-cell renal cancer. Somatic or germline inactivating mutations in the tumour suppressor gene von Hippel–Lindau (*VHL*), a master regulator of hypoxia-inducible factors (HIFs) and the hypoxia response, are found in most clear-cell renal-cell carcinoma (ccRCC) cases (somatic mutations reported in >55% of sporadic ccRCC)^{45,46}. However, studies from model organisms have pointed to the requirement for additional genetic alterations to drive tumour development. Indeed, focused Sanger-based sequencing revealed mutations in neurofibromin 2 (*NF2*) and in genes involved in histone methylation and demethylation:

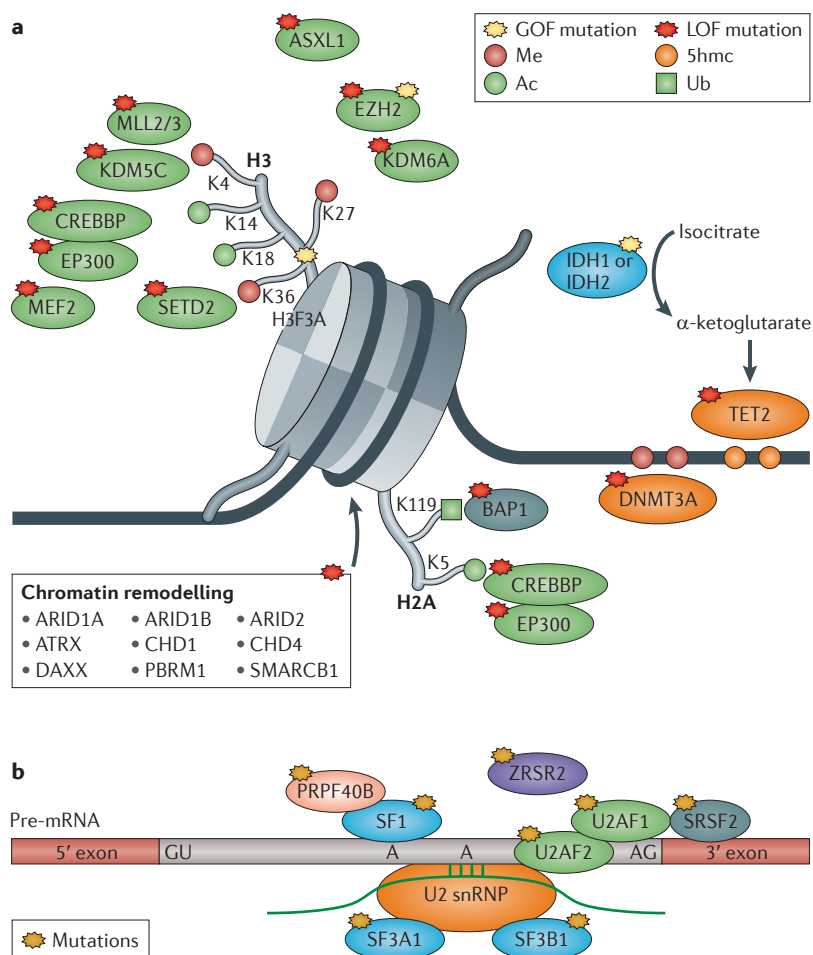


Figure 1 | Frequent mutations of epigenetic regulators and pre-mRNA splicing machinery in cancers. Representative proteins involved in chromatin remodelling, histone and DNA methylation (Me), histone acetylation (Ac) and ubiquitylation (Ub) (part a), and pre-mRNA splicing (part b) that were identified to be recurrently mutated in various types of cancers by whole-exome and whole-genome sequencing are displayed. The predicted consequences of mutations, either gain of function (GOF) or loss of function (LOF), are also shown. 5hmC, 5-hydroxymethylcytosine; ARID, AT-rich interactive domain; ASXL1, additional sex combs like 1; ATRX, α -thalassaemia/mental retardation syndrome X-linked; BAP1, BRCA1-associated protein 1; CHD, chromodomain helicase DNA-binding protein; CREBBP, cAMP responsive element binding protein (CREB)-binding protein; DAXX, death-domain-associated protein; DNMT3A, DNA methyltransferase 3A; EP300, E1A-binding protein p300; IDH, isocitrate dehydrogenase; KDM5C, lysine-specific demethylase 5C; KDM6A, lysine-specific demethylase 6A; MEF2, myocyte enhancer factor 2; MLL, mixed-lineage leukaemia; PBRM1, polybromo 1; PRPF40B, PRP40 pre-mRNA processing factor 40 homologue B; SETD2, SET domain-containing 2; SF, splicing factor; SMARCB1, SWI/SNF related, matrix-associated, actin-dependent regulator of chromatin, subfamily b, member 1; snRNP, small nuclear ribonucleic particle; SRSF, serine/arginine-rich splicing factor; TET2, tet methylcytosine dioxygenase 2; U2AF, U2 small nuclear RNA auxiliary factor; ZRSR2, zinc finger (CCCH type), RNA-binding motif and serine/arginine-rich 2.

Table 1 | Overview of example cancer genomics studies for different cancer types

Cancer type	Mutation analysis	Number of samples	Highlighted or novel mutated genes	Refs
Glioblastoma	PCR amplified and sequenced 20,661 genes	22	<i>IDH1</i> ^{**}	6
	WES of paediatric glioblastoma	48; 6 matched normal DNA	<i>H3F3A</i> ^{**} , <i>ATRX</i> [§] , <i>DAXX</i> [§] and <i>TP53</i>	118
Clear-cell renal-cell carcinoma	WES	7	<i>PBRM1</i> [§]	47
	WGS	1	<i>BAP1</i> [§]	53
	WES	106	<i>TCEB1</i> ^{**§} , <i>KEAP1</i> [§] , <i>TET2</i> and <i>MTOR</i> [†]	55
	WGS	14		
	WES	417	<i>TCEB1</i> ^{**§} , <i>NFE2L2</i> [†] , <i>ARID1A</i> [§] and <i>MTOR</i> [†] (confirmed <i>PBRM1</i> [§] , <i>SETD2</i> [§] , <i>KDM5C</i> [§] , <i>PTEN</i> [§] , <i>BAP1</i> [§] , <i>TP53</i> and <i>PIK3CA</i> ^{**†})	19
	WGS	22		
Head and neck squamous cell carcinoma	WES	32	<i>NOTCH1</i> [§]	56
	WES	74	<i>NOTCH1</i> [§]	57
High-grade serous ovarian carcinoma	WES	316	<i>TP53</i> (universal)	25
Melanoma	WGS	25	<i>PREX2</i> (BOX 4)	61
	WES	121	<i>RAC1</i> ^{**†} , <i>PPP6C</i> [*] , <i>STK19</i> [*] and <i>ARID2</i> [§]	43
	WES	147; 99 matched normal DNA	<i>RAC1</i> ^{**†} , <i>PPP6C</i> ^{**§} and <i>ARID2</i> [§]	62
Lung adenocarcinoma	WES	159	<i>U2AF1</i> ^{**†} , <i>RBM10</i> [§] and <i>ARID1A</i> [§]	64
	WES and WGS	23		
	WGS	1		
Lung squamous cell carcinoma	WES	178	<i>HLA-A</i> [§] , <i>NFE2L2</i> [†] and <i>KEAP1</i> [§]	26
Small-cell lung cancer	WES	53	SOX family (plus 22 SMGs identified)	65
Prostate cancer	WES	112	<i>SPOP</i> , <i>FOXA1</i> and <i>MED12</i> [*]	68
	WES	50	<i>CHD1</i> [§] , <i>MLL2</i> and <i>FOXA1</i> [†]	69
Colorectal cancer	WES	224	<i>ARID1A</i> [§] , <i>SOX9</i> [§] and <i>FAM123B</i> [§]	22
	WES	72	TET family, <i>ERBB3</i> [†] and <i>ATM</i> [§]	70
	WGS	2		
Gastric cancer	WES	22	<i>ARID1A</i> [§]	34
	WES	15	<i>FAT4</i> [§] , <i>ARID1A</i> [§] , <i>MLL3</i> and <i>MLL</i>	35
Breast cancer	WES	54	Confirmed <i>TP53</i> , <i>PIK3CA</i> ^{**†} , <i>RB1</i> [§] and <i>PTEN</i> [§]	75
	WES and WGS	4		
	WGS	15		
	WES	31	<i>MAP3K1</i> [§] , <i>CDKN1B</i> [§] , <i>TBX3</i> , <i>RUNX1</i> , <i>LDLRAP1</i> , <i>STMN2</i> , <i>MYH9</i> , <i>AGTR2</i> , <i>STMN2</i> , <i>SF3B1</i> [*] and <i>CBFB</i> [§]	73
	WGS	46		
	WES	100	<i>AKT2</i> [†] , <i>ARID1B</i> [§] , <i>CASP8</i> [§] , <i>CDKN1B</i> [§] , <i>MAP3K1</i> [§] , <i>MAP3K13</i> [§] , <i>NCOR1</i> [§] , <i>SMARCD1</i> [§] and <i>TBX3</i>	31
	WES	86	<i>CBFB</i> [§]	72
	WES and WGS	17		
	WGS	5		
	WES	510	<i>GATA3</i> [*] , <i>PIK3CA</i> ^{**†} , <i>MAP3K1</i> [§] and <i>MAP2K4</i> [§] (BOX 3)	23
Pancreatic ductal adenocarcinoma	WES	142	<i>EPC1</i> , <i>ARID2</i> , <i>ATM</i> , <i>ZIM2</i> , <i>MAP2K4</i> , <i>NALCN</i> , <i>SLC16A4</i> , <i>MAGEA6</i> and axon guidance pathway genes (<i>SLIT</i> – <i>ROBO</i> signalling)	21
Hepatocellular carcinoma	WES	10	<i>ARID2</i> [§]	37
	WES	24	<i>ARID1A</i> [§] , <i>RPS6KA3</i> [§] , <i>NFE2L2</i> [†] and <i>IRF2</i> [§]	36
	WGS	27	<i>ARID1A</i> [§] , <i>ARID1B</i> [§] , <i>ARID2</i> [§] , <i>MLL</i> [§] and <i>MLL3</i> [§]	27
	WES	10	<i>ARID1A</i> [§]	38

Table 1 (cont.) | Overview of example cancer genomics studies for different cancer types

Cancer type	Mutation analysis	Number of samples	Highlighted or novel mutated genes	Refs
Cholangiocarcinoma	WES	8	<i>TP53</i> , <i>KRAS</i> [‡] , <i>SMAD4</i> [§] , <i>MLL3</i> [§] , <i>ROBO2</i> [§] , <i>RNF43</i> [§] , <i>PEG3</i> [§] and <i>GNAS</i> ^{*†}	39
Acute myeloid leukaemia	WGS	24	<i>SMC3</i> [§] , <i>SMC1A</i> [§] , <i>STAG2</i> [§] and <i>RAD21</i> [§]	88
	WGS	50	Confirmed 23 SMGs: <i>FLT3</i> ^{*†} , <i>NPM1</i> , <i>DNMT3A</i> ^{*§} , <i>IDH1</i> ^{*†} , <i>IDH2</i> ^{*†} , <i>TET2</i> [§] , <i>RUNX1</i> ^{*§} , <i>TP53</i> [§] , <i>NRAS</i> ^{*†} , <i>CEBPA</i> [§] , <i>WT1</i> [§] , <i>PTPN11</i> [†] , <i>KIT</i> ^{*†} , <i>U2AF1</i> , <i>KRAS</i> ^{*†} , <i>SMC1A</i> [§] , <i>SMC3</i> [§] , <i>PHF6</i> [§] , <i>STAG2</i> [§] , <i>RAD21</i> [§] , <i>FAM5C</i> , <i>EZH2</i> ^{*§} and <i>HNRNP</i> ^K	33
	WES	150		
Myelodysplastic syndromes	WES	29	<i>SF3B1</i> [*] , <i>SRSF2</i> [*] , <i>U2AF1</i> [*] , <i>ZRSR2</i> , <i>SF3A1</i> , <i>PRPF40B</i> , <i>U2AF2</i> and <i>SF1</i>	89
	WES	9	<i>SF3B1</i> [*]	28
Chronic lymphocytic leukaemia	WES	5	<i>NOTCH1</i> ^{*†}	94
	WGS	4	<i>NOTCH1</i> ^{*†} , <i>MYD88</i> ^{*†} , <i>XPO1</i> ^{*†} and <i>KLHL6</i>	29
	WES	88	<i>SF3B1</i> [*]	92
	WGS	3		
	WES	105	<i>SF3B1</i> [*]	30
Diffuse large B cell lymphoma	WES	6	<i>MLL2</i> [§] , <i>CREBBP</i> [§] and <i>EP300</i> [§]	100,101
	WES or WGS	13	<i>MLL2</i> [§] and <i>MEF2B</i> ^{*†}	98
	WES	55	<i>MEF2B</i> ^{*†} , <i>MLL2</i> [§] , <i>BTG1</i> , <i>GNA13</i> , <i>ACTB</i> , <i>P2RY8</i> , <i>PCLO</i> , <i>TNFRSF14</i> [§] and <i>BCL2</i> [§]	99
Multiple myeloma	WES	16	<i>DIS3</i> [§] , <i>FAM46C</i> , <i>LRRK2</i> , <i>BRAF</i> ^{*†} , <i>IRF4</i> [†] and 11 NF- κ B pathway genes	103
	WGS	23		
	WES and WGS	1		

ACTB, actin- β ; AGTR2, angiotensin 2 receptor, type 2; ARID, AT-rich interactive domain; ATM, ataxia telangiectasia mutated; ATRX, α -thalassaemia/mental retardation syndrome X-linked; BAP1, BRCA1-associated protein 1; BCL2, B cell chronic lymphocytic leukaemia/lymphoma 2; BTG1, B cell translocation gene 1, antiproliferative; CASP8, caspase 8; CEBF, core-binding factor, β -subunit; CDKN1B, cyclin-dependent kinase inhibitor 1B; CEBPA, CCAAT/enhancer-binding protein- α ; CHD1, chromodomain helicase DNA binding protein 1; CREBBP, cAMP responsive element binding protein (CREB)-binding protein; DAXX, death-domain-associated protein; DNMT3A, DNA methyltransferase 3A; EP300, E1A-binding protein p300; FAM, family with sequence similarity; FAT4, FAT atypical cadherin 4; FLT3, fms-related tyrosine kinase 3; FOXA1, forkhead box A1; GNA13, guanine nucleotide binding protein, α 13; H3F3A, H3 histone family 3A; HLA-A, human leukocyte antigen A; HNRNP^K, heterogeneous nuclear ribonucleoprotein K; IDH, isocitrate dehydrogenase; IRF, interferon regulatory factor; KDM5C, lysine-specific demethylase 5C; KEAP1, kelch-like ECH-associated protein 1; LDLRAP1, low density lipoprotein receptor adaptor protein 1; LRRK2, leucine-rich repeat kinase 2; MAGEA6, melanoma antigen family A, 6; MAP, mitogen-activated protein; MED12, mediator complex subunit 12; MEF2B, myocyte enhancer factor 2B; MLL, mixed-lineage leukaemia; MTOR, mammalian target of rapamycin; MYD88, myeloid differentiation primary response gene 88; MYH9, myosin, heavy chain 9, non-muscle; NALCN, sodium leak channel, non-selective; NCOR1, nuclear receptor co-repressor 1; NFE2L2, nuclear factor (erythroid-derived 2)-like 2; NF- κ B, nuclear factor- κ B; NPM1, nucleophosmin 1; P2RY8, purinergic receptor P2Y, G-protein coupled 8; PBRM1, polybromo 1; PCLO, piccolo presynaptic cytomatrix protein; PEG3, paternally expressed 3; PHF6, PHD finger protein 6; PIK3CA, phosphatidylinositol 3-kinase catalytic subunit- α ; PPP6C, protein phosphatase 6, catalytic subunit; PRPF40B, PRP40 pre-mRNA processing factor 40 homologue B; PTEN, phosphatase and tensin homologue; PTPN11, protein tyrosine phosphatase, non-receptor type 11; RB1, retinoblastoma 1; RBM10, RNA-binding motif protein 10; RNF43, ring finger protein 43; RPS6KA3, ribosomal protein S6 kinase, 90 kDa, polypeptide 3; RUNX1, runt-related transcription factor 1; SETD2, SET domain-containing 2; SF, splicing factor; SLC16A4, solute carrier family 16, member 4; SMARCD1, SWI/SNF related, matrix-associated, actin-dependent regulator of chromatin, subfamily d, member 1; SMC, structural maintenance of chromosomes; SMG, significantly mutated genes; SRSF2, serine/arginine-rich splicing factor 2; STAG2, stromal antigen 2; STK19, serine/threonine kinase 19; TET, tet methylcytosine dioxygenase; TNFRSF14, tumour necrosis factor receptor superfamily, member 14; U2AF1, U2 small nuclear RNA auxiliary factor 1; WES, whole-exome sequencing; WGS, whole-genome sequencing; WT1, Wilms tumour 1; XPO1, exportin 1; ZIM2, zinc finger, imprinted 2; ZRSR2, zinc finger (CCCH type), RNA-binding motif and serine/arginine-rich 2. *Mutations with hotspot identified. †Mutations that are activating or likely to be activating. §Mutations that are inactivating or likely to be inactivating.

Kategoies

Greek for 'shower' or 'thunderstorm'. A phenomenon, identified in breast cancers, of localized hypermutations almost exclusively involving cytosine base pair substitutions at TpC dinucleotides. This mutation pattern has been linked to the APOBEC family of cytidine deaminases.

SET domain-containing 2 (*SETD2*), lysine-specific demethylase 5C (*KDM5C*; also known as *JARID1C*) and *KDM6A* (also known as *UTX*)⁴⁶ (BOX 3; FIG. 1a). A follow-up WES analysis led to the identification of the second most frequently mutated gene in ccRCC, polybromo 1 (*PBRM1*)⁴⁷, which had truncating mutations in 41% of cases (92 of 227). *PBRM1* encodes the chromatin-targeting subunit of the PBAF SWI/SNF chromatin remodelling complex, which regulates transcription and has essential roles in the maintenance of stem cell pluripotency (FIG. 1a) (reviewed in REF. 48). The importance of the PBAF complex was initially suggested from studies that found inactivating mutations in its core component, *SMARCB1* (also known as *SNF5*), in rhabdoid tumours^{49,50}. *PBRM1* mutations were found

in the context of loss of heterozygosity, and functional data supported its role as a tumour suppressor, suggesting that the mutation of *PBRM1* represents the second major genetic event that cooperates with *VHL* in ccRCC⁴⁷. A second WES study identified frequent mutations in the two-hit tumour suppressor BRCA1-associated protein 1 (*BAP1*) in ccRCC, which had been previously found to be frequently mutated in uveal melanoma⁵¹ and pleural mesothelioma⁵², that functions as a deubiquitylating enzyme and regulator of histone H2A lysine 119 ubiquitylation⁵³ (FIG. 1a). Interestingly, mutations in *BAP1* and *PBRM1* were anti-correlated, and tumours with *BAP1* mutations were associated with high tumour grade^{53,54}. Recent efforts from a larger WES and WGS study⁵⁵, and the TCGA¹⁹ confirmed *PBRM1*,

Box 2 | Determining significantly mutated genes and driver mutations

Driver mutations

Somatic mutations in a gene that confer a selective advantage on cancer cells as reflected in the statistical evidence of positive selection. This is not a definition based on functional activity.

Significantly mutated genes

(SMGs). Genes that have a somatic mutation rate above the calculated background mutation rate as determined by a given statistical calculation.

Background mutation rate

(BMR). The rate of mutation in a tumour sample as a consequence of exposure to environmental mutagens (for example, ultraviolet radiation) and/or random generation and misrepair processes.

RNA sequencing

(RNA-seq). Whole-transcriptome shotgun sequencing of cDNA to determine the sequence of RNA; used for expression analysis and the identification of gene–gene fusions.

Two-hit tumour suppressor

Refers to the necessity to inactivate both alleles of a tumour suppressor gene, following the Knudson two-hit hypothesis, which was proposed to explain the early onset of cancer in hereditary syndromes in which the inheritance of one germline copy of a mutated gene in all cells substantially increases the likelihood of any cell acquiring a mutation in the other allele.

Passenger mutations

Neutral mutations in a gene that do not provide a selective advantage for cancer cells as reflected by the lack of statistical evidence for positive or negative selection. This is not a definition based on functional activity.

Hotspot mutation

A recurrent mutation resulting in the same amino acid change in a gene observed in cancer, signifying strong positive selection.

Significantly mutated genes

Several statistical tools to identify significantly mutated genes (SMGs) that have a higher mutation rate than the calculated background mutation rate (BMR) have been developed^{17,131–135}, which include MutSig^{24,26,136}, MuSiC¹³⁷ and InVEx algorithms⁴³. Recent analyses have illustrated how challenging this task is in the context of tumours with a high and heterogeneous BMR^{26,43,64}. Controlling for type I errors (false positives) is difficult, as mutations in a gene can occur in excess of the calculated BMR if they reside in a hypermutable region of the genome (see the figure, part a). Studies have shown that across the genome, mutation rates correlate with the levels of heterochromatin-associated histone H3K9me3 modification, gene expression owing to transcription-coupled repair, the size of genomic footprint of a gene and the replication time of DNA during the cell cycle^{42,103,138–141}. Statistical tests assuming a uniform BMR across the genome may identify genes that are significantly mutated owing to an increased locus-specific BMR and not because of positive selection of the mutated gene in the tumour. SMG lists often include genes that are not expressed in the tumour (for example, olfactory receptors) and extremely large genes (for example, genes involved in muscle development)^{42,65}. Ongoing refinements to algorithms that take such factors into consideration to increase the accuracy of BMR calculations^{42,43} are improving our ability to identify the mutated genes that are undergoing selection. For example, InVEx uses sequencing data from intronic and untranslated regions (UTRs) to infer locus- or gene-specific mutation rates. Based on the assumption that mutations in intronic and UTR sequences are more likely to be under neutral selective pressure, whereas coding mutations will undergo selection during tumour progression (see the figure, part b), InVEx uses a permutation-based framework to determine whether the observed coding mutations experience positive selection against the inferred gene-specific BMR⁴³. Recently, the algorithm MutSigCV⁴² has been developed, which corrects for mutation variation by taking into account patient-specific mutation frequency and spectrum, as well as incorporating expression level and replication time.

WES and WGS analysis of a few samples has demonstrated the ability to discover new ‘mountains’ in cancer, such as the case of polybromo 1 (*PBRM1*) in clear-cell renal-cell carcinoma⁴⁷. However, given that driver mutations can be present at a low frequency in many cancer genes, the International Cancer Genome Consortium calculated that 500 tumour samples are needed to detect, with 80% power, the genes mutated in 3% of samples assuming a typical BMR of 1.5 mutations per megabase^{4,18}. As discussed above, sequencing more samples will not be the only solution, and applying evolving algorithms is equally important.

Distinguishing driver from passenger mutations

A major challenge in the interpretation and translation of genomic discoveries is determining whether SMGs that are prioritized on the basis of statistical significance truly have functional roles in processes that are important for tumorigenesis, and the gene functions that are being modulated by these mutations. There are at least three reasons for a gene to be found as significantly mutated in a sequencing study. First, the identified mutations are technical artefacts. However, in most cases, these genes will be eliminated from the analysis by manual inspection and verification studies. Second, the mutations in a gene are found to occur in excess of the calculated BMR because they reside in a hypermutable region as discussed above (see the figure, part a). Third, a gene is found to be a SMG because it contains driver mutations that are selected during tumorigenesis, and therefore warrants further functional investigation (BOX 4). Additional characteristics can provide supportive evidence of driver mutations. Hotspot mutations and clustered mutations are strong indicators of positive selection, especially when located in functional domains or when they affect amino acid residues that have been shown to be important by three-dimensional protein structures. The same mutations may be found as drivers in other cancers. Evolutionary conservation of a mutated residue may indicate the importance of the mutational event. Algorithms that score mutational impact by amino acid conservation can infer probable functional mutations in cancer genes^{142–144}. Similarly, the tool Cancer-Specific High-Throughput Annotation of Somatic Mutations (CHASM) can help to identify driver mutations by separating candidate driver and passenger mutations on the basis of machine-learning of approximately 50 characteristics associated with cancer-causing mutations¹⁴⁵.

SETD2, *KDM5C* and *BAP1* to be significantly mutated in a cohort of 417 ccRCC tumours, and also identified a novel hotspot mutation in a component of the VHL E3 ligase complex *TCEB1* (REFS 19,55). Importantly, the TCGA study showed widespread DNA hypomethylation in *SETD2*-mutant tumours, and transcriptional network analysis suggests that mutations in the chromatin remodelling complex (*PBRM1*, AT-rich interactive domain 1A (*ARID1A*) and *SMARCA4*) are linked to RAS signalling, immune function, DNA repair, and β -catenin and transforming growth factor- β (TGF β) signalling. RNA-seq analysis also revealed recurrent

a Hypermutable gene locus



b Mutations under positive selection during tumorigenesis



splicing factor proline/glutamine-rich (*SFPQ*)–*TEF3* fusions in five of 416 samples, all five of which lack a *VHL* mutation¹⁹.

Head and neck squamous cell carcinomas. The first global view of the somatic mutations in head and neck squamous cell carcinomas (HNSCCs) was published in two studies in 2011, based on WES characterization of 32 and 74 HNSCC–normal pairs^{56,57}. Epidemiologically, tobacco use, alcohol consumption and infection with human papilloma virus (HPV) are known major risk factors for HNSCC. These studies reported that tumours

Box 3 | Integrative and pathway analysis

The integration of multi-dimensional genomic data has been crucial to our understanding of biologically and clinically relevant subtypes in cancer. An example of insights from integrative analyses is the definition of four transcriptomic subtypes in glioblastoma annotated as proneural, neural, classical and mesenchymal based on gene expression profiles¹⁴⁶. Integration with copy number and mutation data showed that the classical subtype is enriched for epidermal growth factor receptor (*EGFR*) alterations, whereas the mesenchymal subtype contains cases with neurofibromin 1 (*NF1*) loss, and the proneural subtype has specific alterations in platelet-derived growth factor receptor A (*PDGFRA*) and isocitrate dehydrogenase 1 (*IDH1*)¹⁴⁶. Similarly, analysis of promoter methylation alterations in 272 glioblastoma tumours identified a subset of patient samples with characteristic promoter DNA methylation classified as glioma CpG island methylator phenotype (G-CIMP), which is associated with *IDH1* mutations¹⁰. Consistent with previous studies, patients with G-CIMP tumours tended to be diagnosed earlier in life and showed better overall survival. In recent clear-cell renal-cell carcinoma (ccRCC) studies, the integration of mutation and expression data provided insight into the potential mechanisms of novel significantly mutated genes (SMGs) in tumorigenesis. For example, neurofibromin 2 (*NF2*) mutations were found in tumours with a non-hypoxic signature and these mutations were mutually exclusive with von Hippel–Lindau tumour suppressor (*VHL*) mutations, suggesting that mutation in *NF2* is a key driver event in this ccRCC subgroup⁴⁶. Conversely, significant co-occurrence of mutations in epigenetic regulators, SET domain-containing 2 (*SETD2*), lysine-specific demethylase 5C (*KDM5C*) and polybromo 1 (*PBRM1*) were found in tumours with *VHL* mutations or hypoxic expression signatures, supporting the notion that mutations in this class of genes may act as the second hit to the widespread inactivation of *VHL* in ccRCC^{46,47}. The authors of The Cancer Genome Atlas (TCGA) lung squamous cell carcinoma (SQCC) study observed that many of the genes involved in oxidative stress and squamous cell differentiation pathways were frequently altered by mutations and somatic copy-number aberrations, and identified genomic alterations were associated with reported lung SQCC expression signatures (classical, primitive, basal and secretory)²⁶. Similarly, authors in the TCGA breast cancer analysis were able to show expression subtype-associated enrichment for SMGs. Such associations include *GATA3*, high frequency of phosphatidylinositol 3-kinase catalytic subunit- α (*PIK3CA*), and probably inactivating *MAP3K1* and *MAP2K4* mutations in the luminal A expression subtype, enrichment of *TP53* (which encodes p53) and *PIK3CA* mutations in the *ERBB2*-expression subtype, and a high frequency of *TP53* mutations in the basal-like expression tumours.

The recent unbiased genomic characterization of various different cancers has provided a view into the landscape of somatic mutation that has been valuable in establishing correlations with other genetic aberrations, and has provided insights into the driving events in genetic subtypes of cancers that were not well understood. Understanding such a landscape can provide both mechanistic and biological insights into the role of SMGs in a given cancer. Mutual exclusivity of genetic alterations is often seen for somatic alterations that have redundant biological effects in a given pathway. One example is the mutually exclusive mutations in the CUL3–KEAP1 E3 ligase complex and its target substrate nuclear factor (erythroid-derived 2)-like 2 (*NFE2L2*) in the oxidative response pathway in lung SQCC^{26,147}. Based on this principle, the Mutual Exclusivity Modules in cancer method can be used to identify candidate driver networks across a set of patient samples¹⁴⁸.

The usefulness of testing whether known pathways are enriched in SMG lists has been demonstrated in many studies, including the whole-exome sequencing analysis of head and neck squamous cell carcinoma that observed the deregulation of squamous cell differentiation through somatic mutations⁵⁷, and the recent International Cancer Genome Consortium pancreatic study that discovered the enrichment of mutations in genes in the axon guidance pathway²¹. The reverse is also possible: identifying SMG networks using computational strategies, such as HotNet, may help to discover additional pathways that would not have been otherwise identified using more traditional approaches¹⁴⁹.

isolated from patients with a history of tobacco use had higher mutation rates than those from non-smokers, that HPV-associated tumours had far fewer mutated genes, and that mutations in *TP53* (which encodes p53) and HPV infection were mutually exclusive^{56,57}. These observations demonstrate how environmental causes of cancer leave ‘footprints’ in the genome and may offer hints about the molecular mechanism of pathogenesis.

In terms of the genes that are frequently mutated in HNSCC, one study identified six recurrently mutated genes based on the frequency of mutations⁵⁶. Another study defined 39 SMGs using the MutSig algorithm⁵⁷ (BOX 2; see [Supplementary information S1](#) (figure)) and also looked for enrichment in functional gene sets among the list of SMGs; the authors discovered that the highest scoring set was involved in epidermal development, particularly in squamous cell differentiation, which points to the disruption of the stratified squamous differentiation programme as a candidate route to HNSCC.

Both studies identified previously unrecognized *NOTCH1* mutations in HNSCC in approximately 10–15% of samples. Although oncogenic activating *NOTCH1* mutations have been observed in a number of haematological malignancies²⁹, the mutations in *NOTCH1* identified in HNSCC had characteristics that are indicative of loss-of-function (LOF) mutations, hence suggesting a tumour-suppressive role in this cancer type. Phenotypes observed in mice are consistent with the idea that *NOTCH1* can function as a tumour suppressor⁵⁸. γ -secretase inhibitors target the signalling pathway downstream of NOTCH1, but the clinical development of such inhibitors in humans has been halted, partly owing to an increased association with skin cancer risk⁵⁹. This example highlights the context-dependent nature of cancer gene functions that is increasingly being recognized, and the fact that different mutations of the same gene are likely to confer different cancer-relevant biological activities (BOX 4), adding to the complexity of applying genomic information in therapeutic decisions.

CpG island methylator phenotype

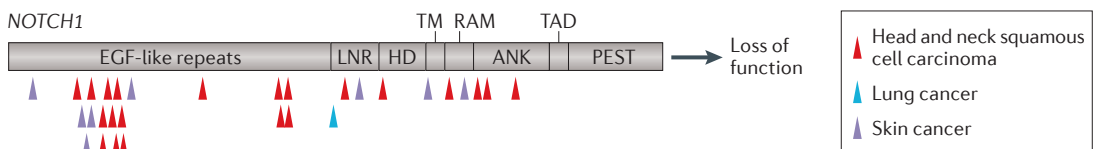
A classification of cancers by their degree of methylation at CpG-rich promoter regions, first characterized in human colorectal cancers; often associated with distinct epidemiological, histological and molecular features.

Box 4 | **Functional validation of significantly mutated genes**

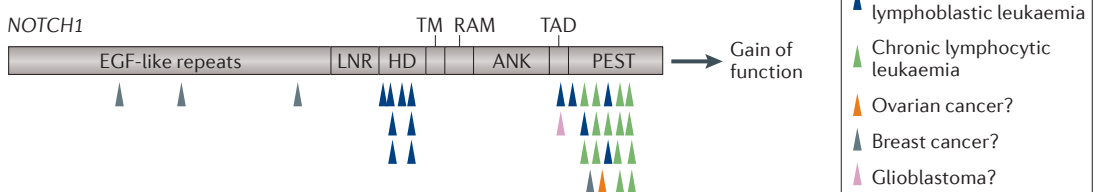
Functional validation of significantly mutated genes (SMGs) and candidate driver mutations requires equal consideration in the selection and interpretation of biological assays used to determine their relevance to cancer. In the case of newly discovered SMGs, it may not be entirely clear in what stage of the tumorigenic process the gene has a role, or how easily the function of a gene can be measured by standard assays. The context and relevant cell lines chosen to carry out experiments are also crucial. The example of *NOTCH1*, which acts as a tumour suppressor with loss-of-function mutations in some tumour types (see the figure, part **a**) and as an oncogene with activating mutations in other tumour types (see the figure, part **b**), highlights the challenges in the functional validation of SMGs and candidate driver mutations that have been identified by cancer genomics. In addition, the genetic context of the biological system used to test the function of an SMG is important. For example, if functional assays are carried out in primary cells that have been immortalized by engineering specific genetic alterations, or in cancer cell lines with inactivation of a tumour suppressor or activation of an oncogene that a candidate SMG may function through, a phenotype may not be observed owing to prior deregulation. Similarly, SMGs may require cooperation with additional cancer-relevant genes to affect tumorigenesis or to bypass oncogene-induced senescence. An example is the lineage-specific melanoma oncogene, microphthalmia-associated transcription factor (*MITF*) — when overexpressed alone, *MITF* did not affect the proliferation of immortalized melanocytes in two- or three-dimensional culture but did affect proliferation with co-expression of the *BRAF* V600E mutant, which co-occurs with *MITF* alterations in a large proportion of melanoma samples¹⁵⁰. Biological assays may also provide insight into the function of the mutations that would not have been accurately predicted by computational approaches. For example, *PREX2* is found to have widespread distribution of missense and truncating mutations from a whole-genome sequencing (WGS) analysis of 25 metastatic melanoma–normal pairs, which indicates a tumour-suppressive function⁶¹. However, missense mutations were found to produce mild phenotypes in tumorigenic assays *in vivo*, whereas truncating mutations were shown to have oncogenic activity that could have been elucidated only through functional experiments. A similar observation was made in the case of a p53-inducible phosphatase-encoding gene, *PPM1D*, in which truncating variants were shown to have gain-of-function effects in functional studies¹⁵¹. The examples discussed above remind us of the need for improved methods to functionally validate candidate cancer genes, such as the development of non-germline mouse models that offer increased speed and reduced costs in developing more high-throughput *in vivo* models¹⁵².

The incorporation of forward genetics has been shown by researchers to aid the determination of the biological relevance of SMGs, as demonstrated by the recent International Cancer Genome Consortium pancreatic study²¹. Cross-species comparative oncogenomics is another approach that can be used to identify driving events. In contrast to cancer genomes from humans, those from genetically engineered mouse (GEM) models have far fewer genetic alterations in addition to the engineered cancer-associated alleles⁴. The presence of homologous, orthologous or paralogous somatic mutations for a given gene in samples obtained from both human and GEM models may indicate strong selective pressure, providing evolutionary evidence that such mutation is a driving event. This approach has been successfully used by a number of groups to identify novel cancer genes targeted by somatic copy-number aberrations in different cancers^{153–155}. More recently, WGS of mouse acute promyelocytic leukaemia (APL) tumours was carried out to identify cooperating somatic mutations in a mouse model expressing a *PML–RARA* fusion oncogene known to initiate APL in mice¹⁵⁶. The authors identified a deletion in lysine-specific demethylase 6A (*Kdm6a*), and a recurrent V657F mutation in janus kinase 1 (*Jak1*). The human *JAK1* V658F mutation has been previously reported in APL, demonstrating that WGS analysis of tumours from GEM models can be used as an unbiased approach for discovering functionally relevant mutations.

a Tumour suppressor



b Oncogene



ANK, ankyrin repeat domain; EGF, epidermal growth factor; HD, heterodimerization domain; LNR, Lin-Notch repeats; PEST, proline–glutamic acid–serine–threonine sequences; RAM, RBP-associated molecule domain; TAD, transactivation domain; TM, transmembrane domain.

Ovarian cancer. Integrated analysis of high-grade serous ovarian adenocarcinoma (HGS-OvCa) by the TCGA included mRNA, miRNA, promoter methylation and DNA copy number analyses from 489 tumours, as well as WES data on 316 of these samples²⁵. Although nine SMGs were identified using two algorithms (MutSig and MuSiC), the landscape of somatic mutations in this cancer type was dominated by the near-universal presence of *TP53* mutations (found in 96% of samples). However, there was also evidence for substantial complexity, with multiple infrequent SMGs being identified (see Supplementary information S1 (figure)). The HGS-OvCa genomes had many somatic copy-number aberrations (SCNAs), with 113 significantly recurrent SCNAs, as defined by the statistical approach GISTIC. This level of SCNAs was greater than that for other studies (for glioblastoma, and colon, breast and lung cancers) reported by the TCGA^{22–26}.

Melanoma. The discovery of the *BRAF* V600E mutation in more than 50% of melanomas in 2002 (REF. 60) and the subsequent development of an inhibitor to treat patients with metastatic disease involving *BRAF* mutation is the proof of concept for genomics-informed personalized therapy. It is now known that an additional 20% of melanomas are characterized by recurrent *NRAS* hotspot mutations; however, the driver mutations in the remaining cases remain poorly understood.

The search for additional driver mutations in melanoma has been complicated by the fact that it has the highest BMR of any cancer sequenced so far^{43,61,62}, which can be almost entirely attributed to the abundance of UV radiation-induced C→T transitions in dipyrimidines. As a result, the identification of SMGs is highly susceptible to discovery cohort bias. To overcome this challenge, statistically powered discovery cohorts of many patients and modified analytical tools that take this high BMR into account were needed to make sense of the mutation data for this tumour type, as demonstrated by two recent studies that analysed WES data of cohorts of >100 tumour–normal pairs^{43,62} (see Supplementary information S1 (figure)). For example, an algorithm called InVEx⁴³ was developed to identify SMGs (BOX 2). This method led to the identification of several novel SMGs, many of which harboured hotspot mutations, a pattern of mutation that signifies strong biological selection. One of these was a hotspot mutation in the Rho GTPase *RAC1*, which was identified in both studies at a frequency of 4–9%^{43,62}. This example illustrates that different numbers of samples and analytical algorithms are required for different tumour types based on the status of the cancer genome.

Lung cancer. Lung cancer is classified into two major histological types: small-cell lung cancer (SCLC) and non-small-cell lung cancer (NSCLC). NSCLC is further divided into squamous cell carcinoma (SCC), adenocarcinoma and large-cell carcinoma subtypes. Substantial progress in personalized treatment for this disease has been made in recent years with the demonstration that epidermal growth factor receptor (*EGFR*)-activating

mutations or gene fusion events involving the anaplastic lymphoma receptor tyrosine kinase (*ALK*) can identify patients with NSCLC who are responsive to treatment with the inhibitors of these tyrosine kinases (reviewed in REF. 63).

Recently, WES and WGS data have provided a more detailed view of the somatic mutational landscape of various lung cancer subtypes, and the studies that generated such data include a combination of WES and WGS analyses of 183 lung adenocarcinoma–normal pairs⁶⁴, an integrated analysis of 178 lung SCCs²⁶ and a WES analysis of 53 SCLC samples⁶⁵. These studies showed that lung cancer has the second highest reported mutation rate after melanoma (mean mutation rate was as high as 12.9 mutations per megabase for smokers), with strong mutation signatures associated with tobacco smoke exposure (that is, G→T transversions and C→T transitions in the context of CpG dinucleotides)^{26,64–66}. In one study, the InVEx algorithm was used to address the challenge of this high mutation rate and 25 SMGs were identified⁶⁴ (see Supplementary information S1 (figure)). The authors found that mutations in U2 small nuclear RNA auxiliary factor 1 (*U2AF1*) (FIG. 1b) and *TP53* negatively correlated with progression-free survival. Together with the frequent mutations in RNA-binding motif protein 10 (*RBM10*), the mutations in *U2AF1* point to a role for RNA splicing deregulation in lung cancer, as seen in various haematological malignancies (FIG. 1b).

A modified version of MutSig²⁶, which took into account gene expression level, GC content, local gene density and local relative replication time, was used to identify ten SMGs in the TCGA study of lung SCC (BOX 2; see Supplementary information S1 (figure)). These included previously unreported LOF mutations in human leukocyte antigen A (*HLA-A*) and widespread *TP53* mutations in nearly all the samples analysed²⁶. Integrative analysis identified pathways that are frequently deregulated in lung SCC, which included the confirmation of the oxidative stress response owing to frequent mutations and SCNAs in the cullin 3 (*CUL3*) and kelch-like ECH-associated protein 1 (*KEAP1*) components of an E3 ubiquitin ligase and its target substrate nuclear factor (erythroid-derived 2)-like 2 (*NFE2L2*) in 34% of samples²⁶ (BOX 3). Finally, the SCLC study took gene expression levels into account in their significance analysis and identified 22 SMGs (see Supplementary information S1 (figure)), including frequent mutations and SCNAs in *SOX* family members that were not previously recognized in SCLC⁶⁵. This series of analyses illustrates the genomically distinct nature of these three subtypes of lung cancer despite their common organ site involvement, which heralds the need for genomic classification to complement traditional histopathological diagnosis.

Prostate cancer. Frequent chromosomal rearrangements of ETS transcription factor genes with androgen-responsive promoters, commonly through *TMPRSS2-ERG* fusions, were discovered as driver events in up to 50% of prostate cancers in 2005 (REF. 67). Recurrent somatic point mutations were thought to have a less important

role in prostate tumorigenesis, until two WES studies shed light on novel SMGs in this disease. An analysis⁶⁸ of WES data from 112 treatment-naïve prostate adenocarcinoma–normal pairs identified 12 SMGs (see Supplementary information S1 (figure)); notably, *SPOP*, a substrate-binding subunit of a cullin-based E3 ubiquitin ligase complex, was mutated in ~13% of samples. A WES analysis⁶⁹ of 50 lethal, heavily pretreated castration-resistant prostate cancers and 11 treatment-naïve, high-grade localized prostate cancers identified nine SMGs. The integration of WES and copy-number data found chromodomain helicase DNA-binding protein 1 (*CHD1*), an ATP-dependent chromatin-remodelling enzyme, to be frequently altered (FIG. 1a). Interestingly, both *SPOP*-mutant and *CHD1*-deleted tumours lacked rearrangements in the ETS gene family, thus demonstrating how the molecular characterization of sizeable tumour cohorts can help to identify novel genetic events that define a molecularly distinct subset of a tumour type.

Colorectal and gastric cancer. A molecular characterization of colorectal carcinoma (CRC) by the TCGA included analysis of WES, DNA copy number, promoter methylation, mRNA and miRNA expression data from >200 samples⁷². Interestingly, the somatic mutation rates varied considerably among the samples studied. Approximately 16% of tumours were designated as hypermutated (tumours that have >12 mutations per megabase). These were characterized by high levels of microsatellite instability (MSI), frequent epigenetic silencing of the DNA mismatch-repair pathway gene *MLH1*, mutations in other mismatch repair genes or the DNA polymerase catalytic subunit *POLE*, providing molecular insights into the underlying causes for the increased mutation rate. Interestingly, the hypermutated samples had few SCNAs and harboured frequent concurrent *BRAF* V600E mutations. After the removal of non-expressed genes, MutSig and prior biological knowledge were used to identify 15 mutated genes in the hypermutated cancer samples and 17 in the non-hypermutated samples that are thought to be important for CRC (see Supplementary information S1 (figure)). The two signature genes of CRC, *TP53* and adenomatous polyposis coli (*APC*), which is an antagonist of the WNT signalling pathway, were found to be more frequently mutated in the non-hypermutated tumours. Across the hypermutated and non-hypermutated groups, a near-universal deregulation (92–97%) of the WNT signalling pathway and the PI3K pathways (~50%) was found. However, deregulation of the TGF β and RTK–RAS signalling pathway was observed more frequently in the hypermutated subtype than in the non-hypermutated subtype.

In another study, WES analysis⁷⁰ of 15 MSI and 57 microsatellite stable (MSS) colorectal cancer samples identified 23 SMGs in the MSS cohort. Interestingly, RNA-seq analysis discovered recurrent gene fusions that are predicted to produce functional proteins involving R-spondin family members, *RSPO2* and *RSPO3*, in 3% (two of 68) and 8% (five of 68) of

samples, respectively. The *RSPO* fusions were found to be mutually exclusive with *APC* mutations, and exogenous expression of plasmids encoding these fusion proteins was shown to activate WNT signalling in a human colon cancer cell line. In a separate WGS study⁷¹, a recurrent fusion of *VTG1A* and *TCF7L2* (which encodes a WNT signalling effector TCF4 transcription factor) was found in three of 97 colorectal cancer samples, and a colorectal carcinoma cell line harbouring the fusion was shown to be dependent on its expression for anchorage-independent growth. These multiple findings of frequent deregulation of WNT signalling is consistent with other evidence of APC as an initiating event in CRC and points to various components of WNT signalling as drivers of this disease, particularly in the hypermutated subtype.

Gastric cancer has a high prevalence in East Asia and WES analysis of >15 gastric tumours was carried out in two studies^{34,35}. Both studies identified cell adhesion and junction organization, and chromatin modification as the most enriched pathways affected by SMGs (see Supplementary information S1 (figure)). Interestingly, a member of the SWI/SNF chromatin remodelling family, *ARID1A*, was found to be more frequently mutated and had decreased expression in MSI (83%) and in Epstein–Barr virus (EBV)-infected MSS (73%) gastric cancers compared with in non-EBV-infected MSS cancers (11%)³⁴. Alterations in *ARID1A* were also predictive of improved disease-free survival, suggesting that the deregulation of the SWI/SNF complex represents a unique mechanism of carcinogenesis associated with a distinct clinical behaviour.

Breast cancer. Clinically, breast cancer is categorized into three basic groups: oestrogen receptor (ER) and progesterone receptor (PR) positive; *ERBB2* (also known as *HER2*) amplified; and triple-negative breast cancer (TNBC), which lacks ER, PR and *ERBB2* overexpression. Recently, various large-scale WES and WGS studies of breast cancer have developed new algorithms to reconstruct the clonal evolution of the tumours (BOX 5), shedding light on the mutational processes that are responsible for the generation of somatic mutations in breast cancer, in addition to identifying the SMGs that correlated with well-established clinically relevant subtypes^{16,23,31,72–75}.

For example, a WES and WGS analysis of 65 TNBC cases⁷⁵ identified six SMGs, confirming *TP53* as the most frequently mutated gene in this subtype. Clonal frequency analysis provided evidence that somatic mutations in *TP53*, PI3K catalytic subunit- α (*PIK3CA*) and phosphatase and tensin homologue (*PTEN*) are clonally dominant in most tumours in which they are found, which is consistent with a founder mutation status role in most, but not all, TNBCs.

Currently, the largest sequencing studies for breast cancer include a WES analysis of 79 ER-positive and 21 ER-negative breast tumours³¹, WES and WGS analyses of 103 and 22 breast tumours from diverse subtypes, respectively⁷², and the TCGA WES analysis of 510 breast tumours from 507 patients²³. Although these studies used

Microsatellite instability (MSI). A hypermutable phenotype caused by germline, somatic or epigenetic inactivation in DNA mismatch repair activity.

Epstein–Barr virus (EBV). A member of the Herpes virus family that is associated with the development of particular forms of cancer.

Triple-negative breast cancer (TNBC). One of the subtypes of breast cancer that is defined by the absence of staining for oestrogen receptor, progesterone receptor and *ERBB2* by immunohistochemistry.

Neoadjuvant aromatase inhibitors

Used to treat patients with oestrogen receptor-positive breast cancer before surgical resection and is applied in cases in which tumour size needs to be reduced for breast-conserving surgery. This treatment is not currently considered as a standard of care and is conducted under clinical trials.

diverse approaches to understand the mutational landscape of breast cancer, they shared remarkable overlap in the SMGs that were identified^{23,31,72} (see Supplementary information S1 (figure)), including nearly all genes that were previously associated with breast cancer, novel SMGs in both the transcription factors *TBX3* (REF. 31) and core-binding factor, β -subunit (*CBFB*)⁷², and a recurrent, potentially druggable *MAGI3*–*AKT3* fusion in 3.4% of samples (eight of 235)⁷².

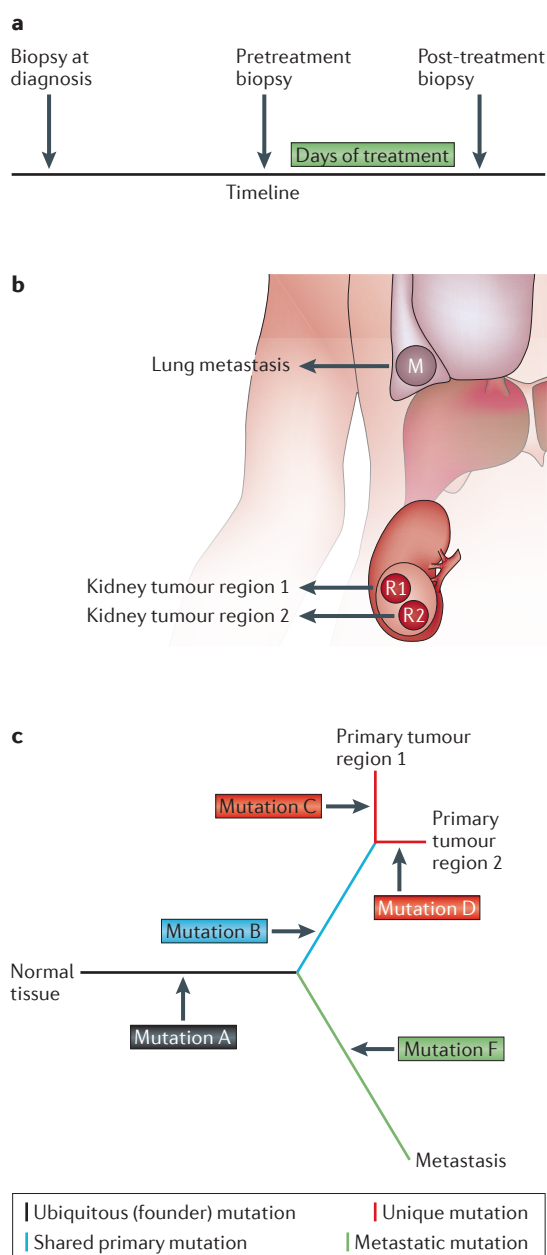
Finally, in a study of the responsiveness of ER-positive breast cancers (of both the luminal A and luminal B expression subtypes) to oestrogen deprivation, WES and WGS analyses were carried out on pretreatment tumour biopsy samples from patients who were

subsequently treated with neoadjuvant aromatase inhibitors⁷³. The authors identified 18 SMGs by MuSiC, and found that *GATA3* mutations correlated with suppression of proliferation in response to aromatase inhibitor treatment. Furthermore, integrative analysis of mutations, mRNA expression and clinical attributes suggested that treatment with neoadjuvant aromatase inhibitors would be a favourable option for patients with *MAP3K1*-mutant luminal A tumours, but not for patients with *TP53*-mutant luminal B tumours⁷³.

Pancreatic cancer. Recently, the ICGC published a WES and copy-number analysis from a prospectively accrued cohort of 142 early (stage I and II) sporadic pancreatic

Box 5 | Role of significantly mutated genes in intra-tumoural heterogeneity and clonal evolution

Several pioneering studies have taken advantage of the sensitivity of next-generation sequencing (NGS) to systematically explore the heterogeneity inherent within the same tumour or the same patient, and how such information can be used to infer the temporal order of the acquisition of genomic alterations (reviewed in REF. 129). Approaches used to measure tumour heterogeneity include mathematical algorithms^{74,130}, single-cell sequencing¹⁵⁷, longitudinal¹⁵⁸ (see the figure, part a) and multi-region sampling^{159,160} (see the figure, part b), in which biopsy samples are taken from different portions of the tumour or from metastatic lesions from the same patient. With such data, one can reconstruct the 'life history' of a tumour by mapping the distribution of clonal and subclonal mutations and somatic copy-number aberrations (see the figure, part c). These analyses can inform whether a mutation in a significantly mutated gene (SMG) is found in the dominant clone, subclones or in a corresponding metastatic tumour, which may imply its prognostic importance and biological role in tumour progression. For example, a whole-genome sequencing analysis from 17 non-small-cell lung cancer–normal pairs validated the presence of epidermal growth factor receptor (*EGFR*) and *KRAS* mutations in founder clones, supporting their roles in cancer initiation¹⁶¹. Studies using longitudinal sampling that involve biopsy samples taken at different time points during the course of a disease have been useful in identifying driver clones that have major roles in disease progression or surviving clones in response to therapeutic regimens by obtaining biopsy samples of cancer cells from patients before and following treatment¹⁵⁸ (see the figure, part a). Recently, geographical sampling and multi-region sequencing of primary renal carcinomas and associated distant metastatic sites demonstrated that 63–69% of all somatic mutations were not detected across every tumour region, and gene expression signatures indicative of both good and poor prognosis were detected in different regions of the same tumour¹⁵⁹. The insights gained from the study design and mathematical tools used in these studies represent a major breakthrough in the analysis of genomic data, and these should be considered and incorporated in all future genomic analyses if possible.



ductal adenocarcinoma (PDAC) samples²¹. MuSiC identified 16 SMGs (see Supplementary information S1 (figure)), and pathway analysis of these genes using the GeneGO algorithm ascertained known cancer pathways (such as the G1/S checkpoint of the cell cycle, apoptosis and TGF β signalling) as mechanisms that are important for PDAC development, as well as a novel pathway involved in axon guidance. Importantly, expression levels of two axon guidance genes, *ROBO2* and *ROBO3*, were associated with patient survival²¹. Forward genetics (in the form of a Sleeping Beauty transposon mutagenesis screen in a mouse model of PDAC) and functional genomics (in the form of a short hairpin RNA screen in pancreatic cell lines) were also used to explore the functional relevance of SMGs that were identified by sequencing. Thus, large-scale genomic analysis coupled with forward genetics and functional genomics can provide insights into pathways that have not been previously linked to cancer.

Liver cancer. HCC has a strong association with chronic liver disease such as viral hepatitis (B or C) and aflatoxin B exposure, and is more prevalent outside of North America and Europe. Four recent genomic studies^{27,36–38} using WES and WGS data from discovery tumour cohorts varying from four to 27 samples have provided insights into the genetic drivers in HCC arising from various aetiologies. All these studies confirmed previously known mutations in *TP53*, but they also shed light on the importance of deregulation by somatic mutations of the genes involved in chromatin remodelling, the WNT- β -catenin pathway, cell cycle control, the PI3K pathway, and oxidative and endoplasmic reticulum stress pathways^{27,36–38} (see Supplementary information S1 (figure)). Although most of the mutations were not associated with a specific type of chronic liver disease, in one study, mutations in interferon regulatory factor 2 (*IRF2*) were exclusively found in hepatitis B virus (HBV)-related tumours³⁶. WGS analysis has also revealed that the number of HBV integration sites in HCC was associated with poor survival, and identified recurrent integration events in the telomerase reverse transcriptase (*TERT*)⁷⁶, mixed-lineage leukaemia 4 (*MLL4*) and cyclin E1 (*CCNE1*) loci, which resulted in a concurrent increase in gene expression^{27,77}, reaffirming the powerful analytical capacity of NGS to investigate the potential role of pathogens in human cancers⁵.

CCA is one form of fatal hepatobiliary cancer that is highly prevalent in certain parts of Southeast Asia where it is associated with the ingestion of the *Opisthorchis viverrini* parasite that is present in raw or undercooked fish. Interestingly, the biliary tree contains stem cell compartments for the liver, pancreas and bile duct, and analysis of WES data from eight *O. viverrini*-related CCA samples identified a mutational landscape that seemed to be more similar to PDAC than to HCC³⁹ (see Supplementary information S1 (figure)). This reinforces the importance of genomic classification in diagnosis, as we begin to understand cancers on a molecular level in addition to their organ site and pathological features.

Landscape of somatic mutations in liquid tumours

Myeloid malignancies. Genomic characterization of haematological malignancies has been at the forefront of the field of cancer genetics, and the most active field in terms of clinical translation. The identification of gene fusions as the predominant drivers of certain leukaemias has led to the development and clinical success of targeted therapies, including imatinib in chronic myeloid leukaemia and acute lymphocytic leukaemia (ALL) cases with the breakpoint cluster region (*BCR*)–*ABL1* fusion and all-*trans*-retinoic acid in acute promyelocytic leukaemia cases with the *PML*–*RARA* (retinoic acid receptor- α) fusion. However, substantial numbers of leukaemias do not possess such gene fusions, and the search for other genetic drivers led to the identification of somatic mutations in genes such as *fms*-related tyrosine kinase 3 (*FLT3*), *RAS*, CCAAT/enhancer-binding protein- α (*CEBPA*), *KIT*, janus kinase 2 (*JAK2*), runt-related transcription factor 1 (*RUNX1*), tet methylcytosine dioxygenase 2 (*TET2*), *ASXL1*, *EZH2* and *TP53* before the era of NGS-based studies^{78–82}. The first WGS analysis of a patient with French–American–British (FAB) classification M1 AML not only confirmed mutations in previously known genes, but also led to the subsequent identification of new somatic mutations in genes that are involved in DNA methylation, such as DNA methyltransferase 3A (*DNMT3A*), *IDH1* and *IDH2* (REFS 83–86) (FIG. 1a).

Although there is some inter-patient heterogeneity, the frequency of somatic mutations that are identified in myeloid leukaemias is generally lower than that of solid tumours. The recent TCGA WES and WGS analysis of 200 patients of *de novo* AML revealed an average of 13 coding mutations per sample (ranging from 0 to 51), and the mutation rate was even lower in cases that harboured known fusion variants, such as the *MLL*–*X* or *PML*–*RARA* fusion³³. In this study, MuSiC identified 23 SMGs that included known mutations in AML, as well as mutations in the mRNA splicing machinery (*U2AF1*) (FIG. 1b) or cohesin complex (structural maintenance of chromosomes 1A (*SMC1A*), *SMC3*, stromal antigen 2 (*STAG2*) and *RAD21*) that were recently identified in myeloid leukaemias^{87,88} (see [Supplementary information S2](#) (figure)).

WES analyses of myelodysplastic syndromes (MDS) have shown that the mutation rate is similar to that of AML (a median of nine mutations per sample), and both diseases share a similar spectrum of mutated genes^{28,89} (see [Supplementary information S2](#) (figure)). However, some genes and pathways are overrepresented in MDS compared with AML and vice versa. For example, mutations in spliceosome complex genes are more abundant in MDS than in AML, and approximately 40% of MDS cases are found to have mutations in one of the spliceosome complex genes (the splicing factors *SF3B1*, *SRSF2*, *SF3A1* and *SF1*, *ZRSR2*, *PRPF40B*, *U2AF1* and *U2AF2*) in a mutually exclusive manner, suggesting that the deregulation in pre-mRNA splicing has a crucial role in the pathogenesis of MDS^{89,90} (FIG. 1b). Recently, sequencing of longitudinal samples from patients with MDS identified hotspot mutations in SET-binding protein 1 (*SETBP1*) that were acquired during the leukaemic evolution to

Sleeping Beauty transposon mutagenesis

A genetically engineered insertional mutagenesis system involving synthetic DNA transposons, which can be applied to various model systems to ascertain gene function.

Aflatoxin B

One of the mycotoxins that are produced by *Aspergillus Flavus*. High-level exposure to aflatoxins is known to cause acute liver necrosis or cirrhosis, resulting in the development of hepatocellular carcinoma.

French–American–British (FAB) classification

First proposed in 1976 by the French–American–British cooperative group and updated in 1989, it classified acute myeloid leukaemia into eight different categories (M0–M7) and acute lymphoblastic leukaemia into three different categories (L1–L3) based on their morphological findings.

AML⁹¹. These data have set the framework for elucidating the genomic basis of transformation from MDS to AML.

Lymphoid malignancies. The pattern of driver mutations identified in lymphoid malignancies differs from that of myeloid malignancies, although there is some overlap of mutated genes found in CLL^{29,30,92–94}, ALL^{95–97}, diffuse large B cell lymphoma (DLBCL)^{98–101}, mantle cell lymphoma (MCL)¹⁰² and multiple myeloma^{103,104} (see Supplementary information S2 (figure)). Examples of findings for CLL include three independent WES and WGS studies^{29,30,92} that revealed known mutations in *TP53* (REFS 105,106) and ataxia telangiectasia mutated (*ATM*)¹⁰⁷, and previously unknown SMGs in *NOTCH1*, myeloid differentiation primary response gene 88 (*MYD88*) and the splicing factor *SF3B1* (REFS 29,30,92) (FIG. 1b). In addition, WES and WGS analysis from 91 CLL cases⁹² defined five core molecular pathways that are crucial for disease pathogenesis: DNA damage repair and cell cycle control, NOTCH signalling, inflammatory pathways, WNT signalling, and RNA splicing and processing pathways.

Some shared features have been reported in other lymphoid malignancies. For example, RNA-seq has identified hotspot mutations in *NOTCH1* in some cases of MCL, suggesting a common role of NOTCH signalling deregulation in B cell malignancies¹⁰². *MYD88* mutations were also observed in DLBCL⁹⁹ and Waldenstrom macroglobulinemia¹⁰⁸. The landscape of somatic mutations has also been characterized in multiple myeloma in two recent WES and WGS studies, confirming known deregulation of RAS, nuclear factor- κ B (NF- κ B) and histone methyltransferase activity, and revealing previously unknown mutations in genes involved in RNA processing and protein homeostasis^{103,104}. The pattern of somatic mutations in DLBCL is more complex⁹⁹, but a number of cases harbour mutations in the regulators of histone and chromatin modification, including *MLL2*, cAMP responsive element binding protein (CREB)-binding protein (*CREBBP*), *EP300* and activating mutation in *EZH2* (REFS 98,100,101) (FIG. 1a). Of interest, WGS analysis of paediatric early T cell precursor ALL revealed a similar somatic mutation landscape to that of myeloid leukaemia, suggesting that therapies effective for patients with myeloid leukaemia might also be effective in this aggressive form of paediatric leukaemia⁹⁷.

NGS also has the potential to reveal major actionable genetic alterations in rare cancers, such as the discovery of *NOTCH2* mutations in 25% of splenic marginal zone lymphoma¹⁰⁹, mutations in signal transducer and activator of transcription 3 (*STAT3*) in 40% of large granular lymphocytic leukaemia¹¹⁰, and *BRAF* V600E mutations present in 100% of hairy cell leukaemia (HCL) samples tested so far¹¹¹. The *BRAF* inhibitor vemurafenib has already shown efficacy in the case of an individual with refractory HCL, and there are ongoing Phase II clinical trials to validate these findings (ClinicalTrials.gov identifier: [NCT01711632](https://clinicaltrials.gov/ct2/show/study?term=NCT01711632))¹¹². These examples show how the systematic integration of data from diverse tumour types has the potential to transform the diagnostic and treatment paradigm for a rare disease.

Genomically defined cancer subtypes

Cancer genomics has shown that histopathologically distinct cancer subtypes of the same organ site often have divergent underlying genomic alterations. In addition to the examples described above, *TP53* is mutated in 96% of HGS-OvCa samples, but clear-cell and endometrioid ovarian tumours have lower rates of *TP53* mutations and instead have frequent recurrent mutations in *PIK3CA*¹¹³ and *ARID1A*^{114,115}. Most cutaneous melanomas are driven by *BRAF* or *NRAS* mutations; by contrast, ocular melanomas have frequent hotspot mutations in the *GNA11* and *GNAQ* genes (which encode G proteins), and LOF mutations in the gene encoding the deubiquitylating enzyme BAP1 (REFS 51,116,117). Recent glioblastoma studies support the notion that paediatric and adult cancers need to be characterized separately at the molecular level, as paediatric and adult glioblastomas have distinct genetic driving events, including mutations in α -thalassaemia/mental retardation syndrome X-linked (*ATRX*), death-domain-associated protein (*DAXX*) and the replication-independent histone variant *H3F3A*, which were much more prevalent in the paediatric setting than in the adult setting¹¹⁸ (FIG. 1a; see Supplementary information S1 (figure)).

Genomically defined cancer subtypes have been shown to have diagnostic and/or prognostic significance. One example of this is the V617F mutation in *JAK2* in the diagnosis of the myeloproliferative neoplasm, polycythemia vera, the incidence of which is estimated at 95% and it is currently incorporated as one of the diagnostic criteria for this condition¹¹⁹. On the basis of recent findings, the *BRAF* V600E mutation might be used as a diagnostic tool for HCL, a disease for which morphological diagnosis has been a challenge, but further validation is required¹¹¹. Classically, genetic information has been actively incorporated into the diagnosis and prognostication of AML and MDS^{120,121}. Chromosomal alterations remain the strongest prognostic factor in both AML and MDS, but recent efforts in the incorporation of somatic mutations have shown promise in creating more sophisticated prognostic models^{122–124}. For example, in intermediate-risk AML identified by the conventional prognostic model, by incorporating information of additional genetic alterations, which include the internal tandem duplication in *FLT3*-ITD and mutations in nucleophosmin 1 (*NPM1*), *CEBPA* and *MLL* genes, physicians are able to identify patients who will benefit from stem cell transplant during the first complete remission¹²⁵. Such prognostication studies help to identify not only biological drivers, but also clinically relevant drivers.

Conclusions and future directions

Several important future directions have emerged from the genomic studies described above. First, the characterization of rare cancers and clinically important genetic subtypes, such as NSCLCs that lack genetic aberrations in *EGFR*, *KRAS* and *ALK*, or melanomas that are wild-type for *BRAF* and *NRAS* will undoubtedly provide valuable information into the genetic aetiology of these cancers. Second, the analysis of somatic

Actionable genetic alterations

Genetic alterations with sufficient scientific evidence supporting their use to inform treatment decisions.

mutations has focused almost entirely on the protein-coding regions of the genome. However, projects such as the Encyclopedia of DNA Elements (ENCODE) are elucidating the functional elements encoded by the roughly 80% of the genome that is non-coding, including promoter and enhancer regions, and are providing an important opportunity to understand the landscape of somatic mutation across the entire genome¹²⁶. In this regard, two groups have recently shown recurrent mutations in the *TERT* promoter in approximately 70% of melanomas^{127,128}, which highlights the discovery potential of examining somatic mutations in non-coding regions of cancer genomes. Third, we need to begin to systematically explore the interaction of host genome variation with the somatic genome of the tumour in ultimately influencing outcomes. Fourth, recent studies

have brought forwards the important issue of tumour heterogeneity and the clinical implications for targeted therapy¹²⁹ (BOX 5). In this respect, in addition to extending genomic study design to include multi-region and longitudinal sampling of tumours, the linkage of complex genomic data to dynamic clinical history of a specific patient will be the most informative in guiding the analyses and in extracting the most clinically relevant insights from the cancer genomes. Such approaches will undoubtedly be essential for the study of drug resistance mechanisms in cancer, which remains an important area of future focus. Finally, we must not underestimate the challenge and necessity of deep biological investigation in the search for efficient and effective translation of new genomic discoveries into end points with clinical impact.

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Competing interests statement

The authors declare no competing financial interests.

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